

Haoran Ding

San Jose, CA · howardding@ucla.edu · (650) 338-7785

EDUCATION

University of California, Los Angeles

B.S. Electrical Engineering, Samueli School of Engineering | GPA: 3.9

Los Angeles, CA

Expected May 2029

Relevant Coursework: CS 31 & CS 32 (Programming in C++ I & II), Math 33A (Linear Algebra), Math 33B (Differential Equations), CS M51A (Logic Design of Digital Systems)

RESEARCH EXPERIENCE

UCLA Summer Undergraduate Research Program (SURP)

Research Fellow — Terahertz and Infrared Photonics Group

Los Angeles, CA

June 2026 – Present

- Developed a MATLAB instrument control suite interfacing the step-scan FTIR spectrometer, Keithley function generator, pulse generator, and lock-in amplifier, consolidating the lab's split LabVIEW/MATLAB workflow into one unified pipeline.
- Automated the end-to-end characterization sequence, biasing the laser and recording optical output across pulsed and continuous-wave operation, eliminating manual instrument reconfiguration and improving repeatability.
- Wrote analysis scripts that ingest acquired datasets to generate L-I-V and emission spectra plots and extract threshold current density, slope efficiency, and characteristic temperature.
- Engineered a MATLAB GUI allowing lab members to configure measurement parameters and run full automated characterization sequences without modifying source code.
- Characterized a previously untested metal-metal waveguide terahertz quantum cascade laser, establishing 2.45–2.61 THz bias tuning and a 133 K maximum operating temperature.

University of California, Santa Barbara

Student Research Associate

Santa Barbara, CA

June 2024 – July 2024

- Engineered bio-inspired snake-scale prototypes to minimize tribological friction, iteratively redesigning interlocking mechanisms to resolve critical 3D-printing tolerance failures.
- Quantified performance using MATLAB and motion-tracking software to extract displacement and velocity datasets, accelerating design iteration cycles.

EXPERIENCE

Handshake AI

AI Training Fellow

Remote

January 2026 – Present

- Evaluate and refine frontier Large Language Models via structured Reinforcement Learning from Human Feedback (RLHF) pipelines, improving model reasoning quality and output accuracy.
- Audit complex AI-generated technical solutions for logical consistency across engineering and CS domains, providing calibrated feedback that informs model fine-tuning.

Bruin Underwater Robotics (ASME) at UCLA

Robotics Engineer

Los Angeles, CA

September 2025 – Present

- Architecting Autonomous Underwater Vehicle (AUV) chassis frame geometry and optimizing thruster configurations for hydrodynamic stability, collaborating with the electrical subsystem team on component integration and system-level design reviews.

MATE ROV — Valley Christian High School

Team CEO

San Jose, CA

September 2022 – May 2025

- Led a multidisciplinary team to design, build, and pilot a Remotely Operated Vehicle, achieving 1st place overall and 1st in Engineering & Communication at the 2023 MATE ROV Monterey Regional.
- Spearheaded CAD modeling of critical ROV components in Onshape, applying hydrodynamic principles and iterative 3D printing to optimize structural weight and drag.

PROJECTS

AI-Powered Market Analyst Agent

Python | Gemini API | Financial Modeling Prep API

2026

- Architected a multi-source financial intelligence pipeline in Python using the Gemini and Financial Modeling Prep APIs, handling edge-case logic to correctly bypass DCF modeling for banking institutions.
- Reduced investment memo generation from hours of manual research to seconds via an end-to-end LLM orchestration workflow producing structured valuation and risk analysis.

Quantum RAG Assistant

Python | ChromaDB | LLM Integration

2026

- Built a Retrieval-Augmented Generation assistant using Python and ChromaDB to ingest, parse, and query complex semiconductor physics research papers, accelerating lab research workflows.

Autonomous Line-Following Robot

C++ | Arduino | PID Control

May 2026

- Implemented a real-time closed-loop PID control algorithm on an Arduino microcontroller, integrating IR sensor arrays with motor driver circuitry and tuning gain coefficients empirically to eliminate steady-state tracking error across variable-contrast surfaces.

TECHNICAL SKILLS

Languages: MATLAB, Python, C++, C, Java

Instrumentation: Instrument Control & Automation, FTIR Spectroscopy, Lock-In Amplifiers, Function & Pulse Generators, Cryostats, LabVIEW, MATLAB GUI Development

AI & Software: Gemini API, REST APIs, RAG, ChromaDB, LLM Prompt Engineering, RLHF

Hardware & Design: Arduino, STM32, PCB Design, Circuit Analysis, Soldering, Onshape (CAD), 3D Printing